

(Also Carroll 2.12)

$$\nabla_\mu \nabla_\sigma K^\rho = \nabla_\mu (\partial_\sigma K^\rho + \Gamma_{\sigma\lambda}^\rho K^\lambda)$$

$$= \partial_\mu (\partial_\sigma K^\rho + \Gamma_{\sigma\lambda}^\rho K^\lambda) + \Gamma_{\mu\nu}^\rho (\partial_\sigma K^\nu + \Gamma_{\sigma\lambda}^\nu K^\lambda) - \Gamma_{\mu\sigma}^\alpha (\partial_\alpha K^\rho + \Gamma_{\alpha\lambda}^\rho K^\lambda)$$

in right
coor., $\partial K = 0$.

$$= \partial_\mu \partial_\sigma K^\rho + K^\lambda \partial_\mu \Gamma_{\sigma\lambda}^\rho + \Gamma_{\sigma\lambda}^\rho \partial_\mu K^\lambda + \Gamma_{\mu\nu}^\rho \partial_\sigma K^\nu + \Gamma_{\mu\nu}^\rho \Gamma_{\sigma\lambda}^\nu K^\lambda - \Gamma_{\mu\sigma}^\alpha \partial_\alpha K^\rho - \Gamma_{\mu\sigma}^\alpha \Gamma_{\alpha\lambda}^\rho K^\lambda$$

$$R^\rho_{\sigma\mu\nu} K^\nu = K^\nu \partial_\mu \Gamma_{\nu\sigma}^\rho - K^\nu \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda K^\nu - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda K^\nu$$

$$\text{equal if } -K^\nu \partial_\nu \Gamma_{\mu\sigma}^\rho = 0.$$

But in right coor., $K^\nu \partial_\nu \rightarrow \partial_{\nu^*}$ where $\partial_{\nu^*} g_{\mu\nu} = 0$. Since Christoffel connection is made of metric terms, this term indeed vanishes.

But $\nabla_\mu \nabla_\sigma K^\rho = R^\rho_{\sigma\mu\nu} K^\nu$ is a tensorial statement, & therefore it holds generally.

$$K^2 \nabla_2 R$$

$$= K^2 \partial_2 R$$

In the right coords,

$$= \partial_{x^2} R$$

R is made of metric terms & Christoffel connections, which are also made of metric terms. As before, this is just zero.

$$\nabla_\mu K_\nu + \nabla_\nu K_\mu$$

$$= \partial_\mu K_\nu - \Gamma_{\mu\nu}^\lambda K_\lambda + \partial_\nu K_\mu - \Gamma_{\nu\mu}^\lambda K_\lambda$$

$$= \partial_\mu (g_{\rho\nu} K^\rho) - \Gamma_{\mu\nu}^\lambda g_{\rho\lambda} K^\rho + \partial_\nu (g_{\rho\mu} K^\rho) - \Gamma_{\nu\mu}^\lambda g_{\rho\lambda} K^\rho$$

$$K^p (\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu})$$

$$= \partial_\mu (g_{\rho\nu} K^p) + \partial_\nu (g_{\rho\mu} K^p)$$

$$- K^p \partial_\mu g_{\nu\rho} - K^p \partial_\nu g_{\rho\mu}$$

$$M^\mu = [K, L]^\mu = K^\lambda \nabla_\lambda L^\mu - L^\lambda \nabla_\lambda K^\mu$$

Show: $\nabla_{(\mu} M_{\nu)} = 0$.

$$\nabla_\mu M_\nu + \nabla_\nu M_\mu$$

$$= \nabla_\nu (K^\lambda \nabla_\lambda L_\mu - L^\lambda \nabla_\lambda K_\mu) + \nabla_\mu (K^\lambda \nabla_\lambda L_\nu - L^\lambda \nabla_\lambda K_\nu)$$

$$= (\nabla_\lambda L_\mu) \nabla_\nu K^\lambda + K^\lambda \nabla_\nu (\nabla_\lambda L_\mu)$$

$$- (\nabla_\lambda K_\mu) \nabla_\nu L^\lambda - L^\lambda \nabla_\nu (\nabla_\lambda K_\mu)$$

$$+ (\nabla_\lambda L_\nu) \nabla_\mu K^\lambda + K^\lambda \nabla_\mu (\nabla_\lambda L_\nu)$$

$$- (\nabla_\lambda K_\nu) \nabla_\mu L^\lambda - L^\lambda \nabla_\mu (\nabla_\lambda K_\nu)$$

$$\text{Ofc, } \nabla_\nu K_\mu + \nabla_\mu K_\nu = \nabla_\nu L_\mu + \nabla_\mu L_\nu = 0.$$

$$\Rightarrow \nabla_\nu K^\mu = -\nabla^\mu K_\nu, \text{ same for } L.$$

$$\begin{aligned} \text{So } (\nabla_\lambda L_\nu)(\nabla_\mu K^\lambda) \\ &= (\nabla_\nu L_\lambda)(\nabla^\lambda K_\mu) \\ &= (\nabla_\nu L^\lambda)(\nabla_\lambda K_\mu) \end{aligned}$$

& similarly:

$$-(\nabla_\lambda K_\nu)\nabla_\mu L^\lambda = (\nabla_\lambda L_\mu)(\nabla_\nu K^\lambda)$$

So:

$$\begin{aligned} \nabla_\mu M_\nu + \nabla_\nu M_\mu &= K_\lambda (\nabla_\nu \nabla^\lambda L_\mu) \\ &\quad - L_\lambda (\nabla_\nu \nabla^\lambda K_\mu) \\ &\quad + K_\lambda (\nabla_\mu \nabla^\lambda L_\nu) \\ &\quad - L_\lambda (\nabla_\mu \nabla^\lambda K_\nu) \\ &= -K_\lambda \nabla_\nu \nabla_\mu L^\lambda + L_\lambda \nabla_\nu \nabla_\mu K^\lambda \\ &\quad - K_\lambda \nabla_\mu \nabla_\nu L^\lambda + L_\lambda \nabla_\mu \nabla_\nu K^\lambda \end{aligned}$$

$$= -K^\lambda R_{\lambda\mu\nu\sigma} L^\sigma + L^\lambda R_{\lambda\mu\nu\sigma} K^\sigma \\ - K^\lambda R_{\lambda\nu\mu\sigma} L^\sigma + L^\lambda R_{\lambda\nu\mu\sigma} K^\sigma$$

Now, $R_{\lambda\nu\mu\sigma} = R_{\mu\sigma\lambda\nu} = R_{\sigma\mu\nu\lambda}$, so:

$$\nabla_{(\mu} M_{\nu)} = -R_{\lambda\mu\nu\sigma} K^\lambda L^\sigma + R_{\lambda\mu\nu\sigma} K^\sigma L^\lambda \\ - R_{\sigma\mu\nu\lambda} K^\lambda L^\sigma + R_{\sigma\mu\nu\lambda} K^\sigma L^\lambda \\ = 0.$$

Relabeling the dummy indices makes it more obvious.